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Beyond Spatial Accuracy: Diagnosing Persistent Orientation Failures in Vision–Language Models

Anonymous WACV Datasets Track submission

Paper ID *****

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Supplementary Material

1. Exact Prompts, Decoding, and Parsing

VSR plain prompt (identical for 2B and 7B, all conditions; '{statement}' is the VSR caption):

```
Look at the image carefully.

Statement: "{statement}"

Is this statement true or false?

Answer with exactly one word:
True or False.
```

2B structured-decomposition prompt (2B only; single run):

```
Look carefully at the image.

Statement:
"{statement}"

Determine whether the statement is true
by doing the following:
1. Identify the subject object.
2. Identify the reference object.
3. Determine the position and
   orientation of the subject relative
   to the reference object.
4. Pay careful attention to left/right,
   front/back, viewpoint, and
   orientation.
5. Decide whether the stated spatial
   relationship matches the image.

Answer with exactly one word:
True or False.
```

SITE multiple-choice prompt (replicated from the official lmms-eval task sitebench/utils.py):

```
# image examples (one <image> token per visual):
<image>...
Question: {question}
Options:
A: {opt0}
B: {opt1}
...
Give me the answer letter directly.
The best answer is:

# video examples (pre-prompt):
Select the best answer to the following
multiple-choice question based on the
video. Respond with only the letter
of the correct option.
Question: {question}
Options:
A: ...
B: ...
...
Give me the answer letter directly.
The best answer is:
```

Decoding. All generations are greedy: do_sample=False, temperature 0, top-p 1.0. max_new_tokens: 5 for the 2B wrapper, 128 for 7B VSR runs, 16 for SITE (change validated parse-identical on 125/125 held-out outputs before adoption; recorded in the SITE run metadata).

MiMo-V2.5 protocol (external-validation row). MiMo-V2.5 (XiaomiMiMo/MiMo-V2.5, 310B total / 15B activated, dedicated 729M vision encoder) is evaluated zero-shot through its public API with the identical frozen VSR prompt above, thinking mode disabled (chat_template_kwargs.enable_thinking=false and thinking.type=disabled; verified content-only, zero reasoning tokens, deterministic on re-runs), temperature 0, top-p 1.0, max_tokens=64, one image per request (canonical COCO URLs, with automatic base64 retry fallback from the local cache), and 12 parallel workers (within the provider's rate limits). Outputs use the same parser; 3.1% of responses did not conform to the one-word format and are counted as wrong (never guessed), and 294 responses contained chain-of-thought text alongside a parseable verdict; both are flagged in the committed predictions (results/mimo/). The evaluation cost is reported in results/mimo/usage_report.json (per-request token accounting).

MiMo-V2.5 output-format sensitivity (additive). Because 69 of the 2,195 outputs (3.1%) were non-conforming, we report the parseable-only counterpart of the headline numbers (results/mimo/mimo_parseable_sensitivity.json): overall accuracy rises from 79.5% (1746/2195, invalid counted wrong) to 82.1% (1746/2126), and orientation is unchanged at 65.7% (90/137) because none of the 69 non-conforming outputs fell in the orientation family. The orientation conclusions are therefore independent of the format-policy choice; the overall row depends on it by 2.6 pp.

Parsing. VSR outputs are parsed with an anchored case-insensitive regex (src/evaluation/parser.py): strips prefixes ('the answer is', 'answer:', 'response:', 'output:') and trailing punctuation; accepts exact 'true/yes/correct' or 'false/no/incorrect' forms; substring fallback only when exactly one of 'true'/'false' appears. Invalid outputs are counted as wrong, never guessed; invalid rate is 0 on all reported runs. SITE outputs use the official parse_multi_choice_response (letter); unparseable responses are counted incorrect and reported.

2. Training and Adapter Configurations

All LoRA conditions share: r=8, α=16, dropout 0.05, AdamW lr 10^{-4}, weight decay 0.01, linear warmup 10% of steps, 2 epochs, bf16, gradient checkpointing, gradient clipping 1.0, vision tower frozen. Adapter configs are committed at checkpoints/*/final/adapter_config.json.

Manifest construction. General: 2,000 statements sampled from the VSR train split. Targeted: orientation-over-sampled subset of the same split. Hard-negative: the orientation block of the general manifest is replaced by audited-clean originals plus paired hard negatives (facing↔facing away from; parallel↔perpendicular) with flipped labels; every included example was audited for label correctness (451 train orientation examples audited: 402 clean, 28 excluded, 21 original-only). 2B runs split 5% of the manifest for validation (seed 42). All manifests derive exclusively from the VSR train split; all evaluations use test only.

3. Probe Methodology and Full Tables

Features. Qwen2-VL-7B-Instruct, vision tower frozen. ViT patch embeddings (1280-d) and post-merger features (3584-d), mean-pooled per image (ungrounded) or per grounding box (grounded). Grounding boxes from frozen Qwen2-VL grounding (94.6% of 647 orientation images yield both boxes; failures are mostly genuinely absent objects), rescaled to pixels; region features mean-pooled over patch centers inside each box.

Tasks. T1 facing vs. facing away (test n=103, majority 64.2%); T2 parallel vs. perpendicular (n=34, majority 64.7%); T3 4-way (n=137, majority 49.6%). Training on audited-clean VSR train; 5-fold CV; no test examples in training.

Readouts. Linear logistic regression; MLP (256 hidden, ReLU, Adam, early stopping on CV); patch-vote (per-patch linear probe, majority vote over patches); grounded variants with feature sets visual ([subj, ref, subj − ref, subj · ref]), geometry (12-d normalized box features), and visual+geometry.

Ungrounded results (test acc %; balanced acc in parentheses; Wilson 95% CI; majority baselines: T1 64.2%, T2 64.7%, T3 49.6%):

T1 facing vs. away (n=103)	Test	Bal. acc	95% CI
ViT linear	65.0	60.4	[55, 74]
ViT MLP	55.3	52.0	[45, 65]
Merger linear	68.9	64.5	[59, 77]
Merger MLP	52.4	49.2	[42, 62]
ViT patch-vote	62.1	—	[53, 71]
Merger patch-vote	60.2	—	[51, 69]

T2 parallel vs. perpendicular (n=34)	Test	Bal. acc	95% CI
ViT linear	61.8	61.0	[44, 77]
ViT MLP	67.6	69.3	[50, 82]
Merger linear	64.7	61.4	[47, 79]
Merger MLP	64.7	63.3	[47, 79]
ViT patch-vote	73.5	—	[57, 85]
Merger patch-vote	67.6	—	[51, 81]

T3 4-way (n=137)	Test	Bal. acc	95% CI
ViT linear	52.6	40.5	[44, 61]
ViT MLP	52.6	42.9	[44, 61]
Merger linear	53.3	44.0	[45, 61]
Merger MLP	47.4	43.2	[39, 56]
ViT patch-vote	51.8	—	[44, 60]
Merger patch-vote	52.6	—	[44, 61]

Grounded results (CV acc / test acc; majority T1 63.7%, T2 57.0%; visual = [subj, ref, subj−ref, subj·ref]; geometry = 12-d box features; T3 4-way is at chance for every readout, test acc 41.7–52.8% across all 12 cells vs. 49.9% majority, committed JSON):

Readout	T1 visual	T1 geom.	T2 visual	T2 geom.
ViT linear	59.8/61.6	61.1/60.6	51.2/53.6	44.2/50.0
ViT MLP	59.5/66.7	61.8/62.6	45.4/46.4	45.4/46.4
Merger linear	57.2/65.7	61.1/60.6	51.2/53.6	44.2/50.0
Merger MLP	56.3/52.5	61.1/71.7	49.0/53.6	50.0/42.9

The single above-majority cell (geometry-only merger MLP on T1, 71.7% test) has CV 61.1% below the 63.7% majority — not robust — and derives from the grounding model's own box statistics, not vision-feature content. In short, object-box conditioning with the model's own grounding predictions and mean-pooled region features does not recover robust orientation decodability under the tested readouts. The T2 patch-vote result (73.5%) is a suggestive point estimate with a wide CI overlapping the majority baseline (n=34).

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Condition	LoRA targets	Trainable	Batch	Manifest
2B General	LM q/k/v/o	∼2.3M	microbatch 4×4	general (2,000)
2B Targeted	LM q/k/v/o	∼2.3M	same	targeted (2,000)
7B LM-only	LM q/k/v/o	∼0.3M	1	general (2,000)
7B HardNeg	LM q/k/v/o	∼0.3M	1	hard-negative (2,000)
7B Projector	visual.merger.mlp.0/2	152K	1	general (2,000)
7B Vision+Proj	merger + blocks 24–31	1.46M	1	general (2,000)

Table 1. LoRA conditions and their target modules.

Clear-subset control (canonical frozen snapshot). The 48-example automated (LLM-generated) audit identified ambiguous or questionable orientation labels; its reliability is assessed against two independent human re-audits in §6. Re-evaluation on progressively stricter subsets (values % from results/tables/orientation_clean_label_table.md, generated by scripts/clean_label_orientation.py): Because auditing changes the evaluated sample, strict-subset values are not a matched comparison against other relation families. The earlier probe-side clear/strict check (pre-freeze) additionally suggested that removing audit-flagged examples does not change the probe conclusion: the best ungrounded probe moved only 68.9% → 71.1% → 71.6% (full/clear/strict, n = 103/97/88) and the generative T1 readouts 64.1/66.0/68.2 (zero-shot) and 68.9/72.2/78.4 (General LoRA); committed in results/probe/clear_subset_results.json. The reliability of the automated audit underlying this table is assessed in §6 (human re-audits + inter-source agreement); the frozen table above is retained unchanged.

Human re-audit tables (versioned, additive). The exclusion masks of Table 2 were originally produced by an automated (LLM-generated) audit whose image grounding is unverified. Two independent human annotators re-rated all 137 orientation examples (binary clean/ambiguous flag) and the 48 persistent failures (eight-class taxonomy) from the images alone under a blind protocol. Applying the same taxonomy-derived strict mask to each human's labels reproduces the qualitative conclusions, with best scores close to the automated one:

Under the human strict masks, orientation accuracy rises (best 77.6% for human A, 78.8% for human B, both 2B Targeted LoRA), error is not eliminated (the best human-mask score 78.8%, n=104, is on par with the automated 78.5%, n=107), and 2B structured still shows no gain (51.4% and 51.0%), so the paper's sensitivity conclusions are not an artifact of the automated audit. The human binary masks alone do not reproduce the gains (best 66.7% human A / 66.2% human B vs. the 66.4% full-set best), which we read as further evidence that clean-label judgments are labeler-dependent at the per-item level; per-item agreement and a consensus analysis are in §6.

4. Complete Statistical Test Battery

All pairwise tests are exact McNemar tests (binomial test on discordant pairs, two-sided). Discordant counts are reported as (condition-loses / condition-gains) relative to the control named in each row. No test was run and withheld; the battery below is complete for this paper.

Principal vs. exploratory comparisons. The battery above is fully enumerated; no test was run and withheld. The principal (confirmatory) comparisons are: zero-shot vs. LM-only consistency (ID 19) and the SITE transfer tests (§E). All other comparisons — vision-side conditions and their per-relation cells, hard-negative training vs. control, the remaining consistency comparisons, and the two-stage agreement rows (IDs 15–18) — are exploratory diagnostics reported with exact unadjusted p-values. Under a Holm adjustment over the 18 accuracy-valid VSR tests (α=0.05), only the zero-shot-vs-LM-only consistency test survives (Holm p=0.0018, using the conservative p<0.0001 bound); the vision-side overall effects (raw p=0.0043, p=0.0122; Holm 0.073, 0.195) and all other comparisons do not survive a blanket adjustment and are not load-bearing. For the 3 SITE transfer tests, Holm-adjusted p-values are 0.012 (primary), 1.0, 1.0.

Single-checkpoint note. Every trained condition is a single checkpoint. All McNemar tests above quantify paired disagreement on the fixed test examples between two conditions; they do not estimate training-run variability, and the single-seed results reported here do not estimate training-run variability. Adapter-rank claims (e.g., the 1–2pp overall differences among trained 7B conditions) should be read in that light. A multi-seed extension (two additional seeds for 7B General LoRA and 7B Hard-Neg LoRA) is prepared in scripts/run_seed_variance.py and scripts/aggregate_seed_variance.py and requires a ≥40 GB GPU; its results, when available, will be reported additively.

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Table 2. Clean-label orientation robustness (canonical frozen snapshot; values % from results/tables/orientation_clean_label_table.md).

Condition	Full (137)	−Questionable (132)	Clear (124)	Strict (107)
2B zero-shot	62.8	65.2	66.9	75.7
2B structured	53.3	52.3	51.6	53.3
2B General LoRA	62.0	63.6	64.5	72.9
2B Targeted LoRA	64.2	65.9	68.5	76.6
7B zero-shot	63.5	65.2	68.5	72.0
7B General LoRA	65.7	67.4	70.2	78.5
7B Targeted LoRA	64.2	65.9	69.4	72.0
7B Hard-Neg LoRA	66.4	68.2	70.2	76.6
7B Projector LoRA	64.2	65.9	69.4	73.8
7B Vision+Projector LoRA	64.2	65.2	68.5	73.8
MiMo-V2.5 zero-shot	65.7	67.4	69.3	71.0

Table 3. Human re-audit clean-label orientation robustness (versioned, additive; values % from results/tables/orientation_clean_label_table_human.md for human A and …_human2.md for human B). The frozen automated-audit table (Table 2) is not modified.

Condition	Full (137)	−Q	Clear	Strict (107)	Human binary (75)
2B zero-shot	62.8	67.5	67.5	76.6	65.3
2B structured	53.3	53.2	53.2	51.4	66.7
2B General LoRA	62.0	66.7	66.7	74.8	66.7
2B Targeted LoRA	64.2	69.0	69.0	77.6	65.3
7B zero-shot	63.5	65.9	65.9	74.8	54.7
7B General LoRA	65.7	68.2	68.2	72.9	66.7
7B Targeted LoRA	64.2	65.1	65.1	69.2	57.3
7B Hard-Neg LoRA	66.4	68.2	68.2	72.9	65.3
7B Projector LoRA	64.2	65.1	65.1	71.0	61.3
7B Vision+Projector LoRA	64.2	67.5	67.5	72.9	64.0
MiMo-V2.5 zero-shot	65.7	68.2	68.2	72.0	58.7

Bootstrap family-relative contrasts (additive). The headline family-relative claim is quantified with a paired bootstrap over the 2,195 shared test examples (10,000 resamples, seed 20260812; results/bootstrap_family_contrast.json, generated by scripts/bootstrap_family_contrast.py). For each family f, Δ_f is the accuracy change between two conditions and the contrast is $\Delta_f - \Delta_{\text{orientation}}$:

The only contrast whose 95% CI excludes zero is the scaling horizontal family (+13.9 pp, positive in 99.3% of resamples); depth and containment are directionally consistent but not individually significant, and under adaptation no family contrast is distinguishable from orientation. The family-relative claim is therefore carried by the horizontal family, and the paper reads the depth/containment differences as weaker supporting evidence rather than independently established.

Consistency verdict patterns. For each strict family and condition, the committed verdict table (results/consistency_verdict_table.json, scripts/consistency_verdict_table.py) reports the observed both-True / both-False / complementary counts, the original and flipped True rates, and the verdict rates expected under verdict independence given those base rates:

The zero-shot pattern is dominated by a False-defaulting response bias: expected consistency under verdict independence is 30.2%, below the observed 36.9%, so a simple base-rate null does not explain the facing inconsistency; conversely, training raises consistency while introducing both-True contradictions (0% → 24.3% LM-only / 17.5% hard-neg), a disclosed specialization trade-off. The large sparse-MoE zero-shot model shows the same False-defaulting bias (0 both-True; 56.0% both-False; observed complementary 44.0% vs. 35.6% expected under independence), so scale alone does not alter the coherence failure mode. Full table for left/right (n=245), front/behind (n=314), and parallel/perp (soft, n=34) in the committed JSON.

Consistency definitions. Strict complements (exactly one truth value): left↔right (n=245), in front of↔behind (n=314), facing↔facing away (n=103). Soft complement (parallel↔perpendicular, n=34): both-False is legitimate; only both-True is a contradiction; both-True rates are reported separately. Self-

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Table 4. Human B (second blind re-audit) clean-label orientation robustness (versioned, additive; values % from results/tables/orientation_clean_label_table_human2.md).

Condition	Full (137)	−Q	Clear	Strict (104)	Human binary (77)
2B zero-shot	62.8	66.4	67.2	77.9	62.3
2B structured	53.3	53.1	52.8	51.0	59.7
2B General LoRA	62.0	65.6	66.4	75.0	63.6
2B Targeted LoRA	64.2	68.8	68.8	78.8	63.6
7B zero-shot	63.5	65.6	66.4	76.0	58.4
7B General LoRA	65.7	68.0	68.8	74.0	62.3
7B Targeted LoRA	64.2	65.6	66.4	71.2	61.0
7B Hard-Neg LoRA	66.4	68.0	69.6	74.0	61.0
7B Projector LoRA	64.2	65.6	64.8	72.1	66.2
7B Vision+Projector LoRA	64.2	67.2	68.0	74.0	63.6
MiMo-V2.5 zero-shot	65.7	68.0	67.2	73.1	62.3

ID	Comparison	Subset	Discord.	p
1	HardNeg vs. Gen.	overall (2195)	52/60	0.508
2	HardNeg vs. Gen.	weak families	20/26	0.461
3	Projector vs. ctrl.	overall	—	0.0043
4	V+Proj vs. ctrl.	overall	—	0.0122
5	Projector vs. ctrl.	orientation	—	0.86
6	V+Proj vs. ctrl.	orientation	—	0.85
7–14	Proj./V+Proj vs. ctrl.	per-relation (4×2)	—	0.25–1.00
15	Two-stage A-lin. vs. ctrl.	agreement (127 boxed)	45/16	0.0003
16	Two-stage A-MLP vs. ctrl.	agreement (127 boxed)	47/14	<0.0001
17	Two-stage B-lin. vs. ctrl.	agreement (127 boxed)	44/20	0.0037
18	Two-stage B-MLP vs. ctrl.	agreement (127 boxed)	47/18	0.0004
19	Zero-shot vs. LM-only	consistency 662	117/43	<0.0001
20	HardNeg vs. LM-only	consistency 662	31/41	0.29
21	Projector vs. LM-only	consistency 662	63/48	0.18
22	V+Proj vs. LM-only	consistency 662	65/56	0.47

Table 5. Complete VSR test battery (22 rows). Per-relation cells (IDs 7–14) are enumerated in the committed vision_side_comparison.json: projector facing p=0.77, facing-away p=1.00, parallel p=0.25, perpendicular p=1.00; vision+proj facing p=1.00, facing-away p=0.58, parallel p=1.00, perpendicular p=1.00. Rows 15–18 are agreement-based: the committed two-stage pipeline scored agreement with the control's verdicts (not accuracy against ground truth), so these rows are descriptive and are excluded from the confirmatory set and from the Holm family. SITE transfer tests (3: all/primary/secondary, §E) form a separate small family.

Table 6. Scaling (2B zs → 7B zs) and adaptation (7B zs → 7B General LoRA) family gains and bootstrap contrasts (pp, 95% CI).

Contrast vs. orientation	Point	95% CI	P(> 0)
Scaling (2B zs → 7B zs)			
horizontal	+13.9	[+2.9, +24.8]	0.993
depth	+5.5	[−5.7, +16.5]	0.834
containment	+5.1	[−5.9, +16.2]	0.820
topology/contact	−0.9	[−11.2, +9.4]	0.429
Adaptation (7B zs → 7B Gen. LoRA)			
depth	+5.0	[−6.5, +16.5]	0.802
topology/contact	+2.0	[−9.6, +13.3]	0.634
containment	+1.5	[−10.1, +13.0]	0.594
horizontal	+0.5	[−10.7, +11.5]	0.538

consistency on a strict pair means opposite verdicts.

Wilson CIs. All per-relation and SITE subset accuracies are reported with Wilson 95% confidence intervals (main text and committed JSONs).

5. SITE Protocol, Keyword Rule, and Distributions

Pre-specification. Protocol frozen 2026-08-09 before any evaluation; example IDs committed in results/site/site_protocol.json; the protocol configuration is recorded in results/site/run_metadata.json. Evaluation-only: no SITE data used for training. Evaluation: Qwen2-VL-7B- Instruct frozen, greedy, official multiple-choice prompt and letter parser, max_new_tokens=16, images resized to ≤392 px long side (constant protocol parameter; the installed transformers build ignores max_pixels), SDPA attention.

Subsets. Primary = official category spatial relationship reasoning (1,721 examples; 993 images). Secondary = heuristic orientation keywords (3,272; 2,024

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Table 7. Facing-pair verdict patterns by condition (n=103); "exp." = expected under verdict independence given the observed base rates.

Condition (facing pairs, n=103)	both-T	both-F	comp.	orig-T	flip-T	exp. both-F	exp. comp.
7B zero-shot	0	65 (63.1%)	38 (36.9%)	0.20	0.17	66.5%	30.2%
MiMo-V2.5 zero-shot (n=100)	0	56 (56.0%)	44 (44.0%)	0.30	0.14	60.2%	35.6%
LM-only LoRA	25 (24.3%)	10 (9.7%)	68 (66.0%)	0.62	0.52	18.0%	49.4%
HardNeg LoRA	18 (17.5%)	5 (4.9%)	80 (77.7%)	0.60	0.52	19.0%	50.2%
Projector LoRA	18 (17.5%)	15 (14.6%)	70 (68.0%)	0.53	0.50	23.5%	50.3%
Vision+Proj LoRA	14 (13.6%)	23 (22.3%)	66 (64.1%)	0.49	0.43	27.1%	49.4%

images). Exploratory = official movement prediction & navigation (1,103; 248 images). Overlaps (image): primary ∩ secondary = 488, secondary ∩ exploratory = 186, primary ∩ exploratory = 0; union = 2,591 images, each evaluated once.

Exact orientation-keyword rule (applied, case-insensitive, word- boundary regex, to question text plus non-image option text; src/datasets/site.py):

orientation, oriented, orient,
facing, face, faces, direction,
directional, view, viewpoint,
view angle, angle, turned, rotate,
rotated, rotation, toward, towards,
away from, left, right, front,
behind, in front of, parallel,
perpendicular, clockwise,
counterclockwise

Secondary subset composition (n=2,024 images) by official category: multi-view & cross-image reasoning 866 (raw 37.8%, CAA 8.8%); spatial relationship reasoning 488 (71.3%, 53.5%); object localization & positioning 346 (46.5%, 28.3%); movement prediction & navigation 186 (49.5%, 24.3%); 3D information understanding 91 (27.5%, 0.4%); counting & existence 47 (53.2%, 37.1%). Top sources: exoego4d 316 (36.4%), egoexo4d 288 (34.7%), MMTBench 237 (48.9%), SAT 183 (46.4%), MMIU 133 (28.6%), CLEVR 106 (92.5%), SPEC 102 (50.0%), SpatialEval 100 (53.0%), ThreeDSRBench 95 (61.1%), MMERealWorld 89 (32.6%), SeedBench 63 (58.7%), GQA 41 (75.6%), VStarBench 37, Blink 35, plus 27 smaller sources.

Decision rule (pre-specified). (1) If the primary subset shows the same broad weakness: run the VSR-trained 7B General LoRA on SITE (pre-specified step 2). (2) Else if only the orientation subset shows the effect: narrow the claim to object/direction orientation. (3) Else: report benchmark-specificity. Outcome: branch 2; the claim was narrowed accordingly. The VSR-trained 7B General LoRA adapter was subsequently evaluated on the identical SITE examples as a post-protocol cross-dataset transfer check (no SITE training; predictions in results/site/vsr_lora_predictions.csv, paired statistics from scripts/compare_site_zeroshot_lora.py, canonical table in results/tables/site_validation_table.md):

Entries are raw accuracy / CAA (%). The adapter is statistically neutral overall, does not improve the orientation subset, and significantly degrades the official spatial-relationship category — VSR gains transfer poorly and can induce negative transfer.

Composition-confound analysis (CPU-only, zero-shot predictions). Because the orientation slice is keyword-derived and heterogeneous, we test whether the heuristic flag predicts lower correctness after controlling for task composition (logistic regression over the 2,591 zero-shot predictions; scripts/site_confound_analysis.py → results/site/orientation_confound_analysis.json):

Within the official spatial relationship reasoning category, orientation-flagged questions remain descriptively 7.5pp harder than unflagged ones (71.3% vs. 78.8%; n=488/505). No source dataset has $n \geq 30$ in both arms (orientation-heavy corpora are almost entirely flagged, others almost never), so source indicators absorb most of the orientation variance in the full model. Reading: the adjusted association is in the expected direction but not significant; the low aggregate heuristic score is largely explained by task composition. We therefore do not treat SITE as independent confirmation of the VSR orientation bottleneck, and the main text reports the slice as exploratory.

High-precision post-hoc subset (exploratory only). Restricting the heuristic to VSR-construct terms only (facing, facing away, parallel, perpendicular; word-boundary match on question or option text) yields n=177 examples (77 in the official spatial-relationship category, 99 in multi-view/cross-image reasoning, 1 in object localization): raw accuracy 44.1% (95% CI $[37.0, 51.4]$), CAA -2.3%. This subset is explicitly post-hoc and does not replace the pre-specified frozen definition; it is reported as exploratory only.

Table 8. SITE VSR-LoRA transfer check (post-protocol; predictions in results/site/vsr_lora_predictions.csv, canonical table in results/tables/site_validation_table.md). Subsets are defined by the frozen protocol IDs (see note below); no keyword reconstruction is used in analysis.

Subset	Zero-shot	VSR-LoRA	Δ	McNemar p
All images (n=2,591)	54.2 / 31.1	53.8 / 30.5	−0.4	0.66
Primary: spatial-rel. (n=993)	75.1 / 59.2	71.6 / 53.4	−3.5	0.004
Secondary: orient. kw. (n=2,024)	48.3 / 24.0	48.7 / 24.5	+0.4	0.70
Exploratory: movement (n=248)	48.8 / 23.1	50.4 / 25.6	+1.6	0.64

Table 9. Composition-confound logistic regressions (2,591 zero-shot SITE predictions; outcome = correct).

Model: correct ∼ orient + controls	OR (orient)	95% CI	p
Unadjusted	0.31	$[0.25, 0.38]$	2.4×10^{-28}
+ official category	0.74	$[0.57, 0.96]$	0.023
+ category, source, modality, $n_{options}$	0.84	$[0.60, 1.16]$	0.29

6. Failure Taxonomy

The 48 persistent-failure statements are the union of: wrong in both 7B zero-shot and 7B LoRA (20); wrong in ≥3 of 4 base conditions (38); wrong in both 2B and 7B zero-shot (28). The taxonomy below was produced by an automated audit (LLM-generated labels; commit 4b83729); its image grounding is unverified, and its reliability is assessed against two independent human re-audits below. The automated audit suggests many persistent errors remain visually interpretable while also identifying viewpoint and annotation ambiguities; it is qualitative evidence, not a causal localization of the failure. Annotations committed in results/failure_annotations.csv and results/orientation_persistent_annotations.csv.

Failure mode	n	Share
Clear image, model reasoning failure	18	37.5%
Camera viewpoint ambiguity	8	16.7%
Parallel/perpendicular geometry	6	12.5%
Annotation questionable (label wrong)	5	10.4%
Intrinsic orientation ambiguous	4	8.3%
Front/back object ambiguous	4	8.3%
Small / occluded object	2	4.2%
Subject/reference inversion	1	2.1%

Table 10. Persistent-failure taxonomy, 48 annotated cases.

Transition counts. Scaling (2B→7B zero-shot): 28 both wrong, 23 2B-wrong/7B-right, 22 2B-right/7B-wrong, 64 both right. Adaptation (7B zero→7B LoRA): 20 both wrong, 30 fixed, 27 broken, 60 both right. Of the 48 annotated cases, hard-negative training fixes 5; 15 still fail under all three 7B conditions. Facing/facing-away account for 68.8% of persistent failures.

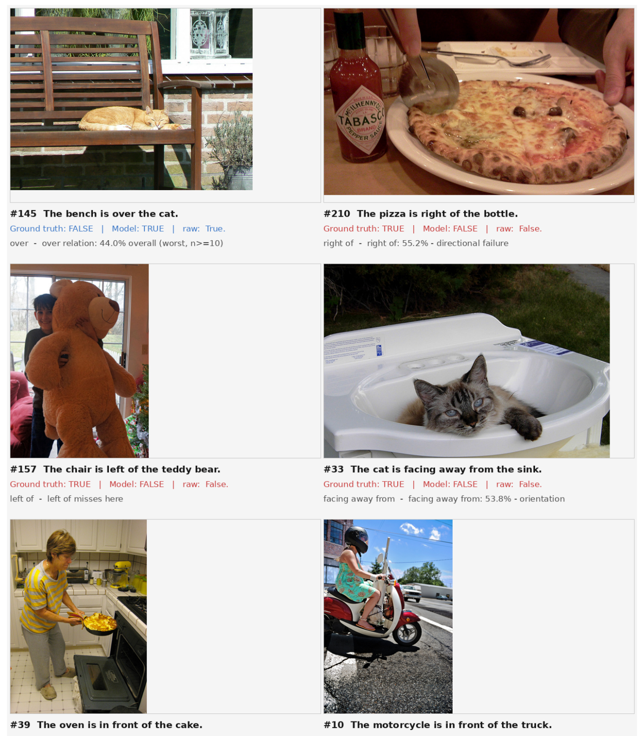

Figure 1. Representative persistent failures (annotated grid; 15 cases). Images: VSR test set. Per the automated audit (corroborated by the human re-audit below), the most frequent class is clear-image reasoning failure: the pose is visually interpretable to a human and the model still errs.

Human re-audit and inter-source agreement (additive reliability evidence). Because the clean-label audit (§3) and the taxonomy above originate from an automated (LLM-generated) audit, two independent human annotators re-rated the same examples from the images under a blind protocol (rater slots in scripts/iaa_tool.py): scripts/export_iaa_sheets.py exports two rating sheets (the 137-example orientation test set for the binary clean/ambiguous flag, and the 48 persistent-failure cases for the eight-class taxonomy) containing only image, statement, and relation—ground truth, model predictions, and the automated audit's labels are withheld—and scripts/iaa_tool.py provides a blind annotation interface with auto-save. Agreement is computed by scripts/compute_iaa.py (Cohen's kappa for the binary flag, Krippendorff's alpha, nominal, for the taxonomy, both with bootstrap 95% confidence intervals; results/iaa/iaa_results.json).

Human–human agreement. The two human annotators agree substantially: on the full 137-example binary flag, 82.5% agreement (κ=0.65, CI $[0.51, 0.77]$); on the 48-case audited subset, 70.8% agreement (κ=0.44, CI $[0.21, 0.67]$). On the eight-class taxonomy the exact-class agreement is 85.4% (Krippendorff's α=0.81, CI $[0.68, 0.92]$; committed at results/iaa/rater2_rater3_*). The binary clean/ambiguous judgment is therefore the fuzzier of the two tasks, while the failure taxonomy is reliably rateable by humans.

Results (human vs. automated audit). On the 48 cases rated by both sources (the persistent-failure set; the remaining 89 orientation examples were never flagged by the automated audit and are excluded from the agreement statistics), the binary clean/ambiguous flag agrees on 32/48 cases (66.7%; κ=0.36, bootstrap 95% CI $[0.13, 0.59]$). On the eight-class taxonomy the exact-class agreement is 15/48 (31.2%; α=0.14, CI $[-0.03, 0.28]$). The disagreement is systematic rather than random: the human annotators never used subject/reference inversion (one used camera-viewpoint ambiguity only three times), over-used annotation questionable (11 and 9 vs. 5) and small/occluded object (11 and 13 vs. 2), and flagged 62 and 60 of the 137 orientation examples as ambiguous, versus 30 excluded by the automated audit's strict set. Part of the disagreement on annotation questionable is attributable to the blind protocol itself: judging whether a ground-truth label is wrong requires the label, which was withheld.

Consequences. The frozen automated-audit table (Table 2) is retained unchanged; we do not redefine the exclusion sets to manufacture agreement. Both human re-audits reproduce the qualitative conclusions with their own taxonomy-derived strict masks (Table 3, human A: best 77.6%, n=107; Table 4, human B: best 78.8%, n=104; both leave 2B structured prompting without gain at 51.4%/51.0%), so the sensitivity result is robust at the aggregate level across two independent humans despite the automated audit's low per-item agreement with either of them. An annotator-dependence analysis (results/iaa/consensus_subsets.json, scripts/iaa_consensus_subsets.py) shows the same pattern: under the humans' binary masks the clean-subset gains largely vanish (best 66.7% human A / 66.2% human B vs. the 66.4% full-set best), while under the consensus union mask (n=62) the trained conditions again improve (7B General LoRA 77.4%, 7B Hard-Neg 74.2%, 2B zero-shot 72.6%) but 7B zero-shot does not (58.1%). The statement that 2B structured prompting shows no clean-subset gain (53.3% on both full and strict sets) is specific to the taxonomy-derived masks: under the human binary consensus mask it reaches 69.3%. Overall the human re-audits provide partial but aggregate-level corroboration, so the clean-label analysis should be read as annotator-specific exploratory evidence rather than a ruling on label noise. Human raters support the aggregate sensitivity conclusion, but per-item exclusion decisions are annotator-dependent, and automated auditors vary substantially in calibration (below). No reported accuracy number is altered by this protocol.

LLM audit reliability is itself model-dependent. Three additional image-grounded LLM auditors ran the same audit prompts (results/iaa/gpt56_clean_137.csv, results/iaa/gpt56_taxo_48.csv, results/iaa/gemini_clean_137.csv, results/iaa/gemini_taxo_48.csv; agreement statistics in results/iaa/model_auditor_agreement.json). MiMo-V2.5's labels collapsed to a single dominant class (43/48 taxonomy cases and 132/137 binary flags), yielding no agreement with either the automated first-pass labels or either human re-audit (κ=0.00; $\alpha \approx -0.1$; results/iaa/human_mimo_results.json, results/iaa/llm_mimo_results.json). GPT-5.6 Sol produces a non-degenerate partition close to the humans' CLEAN base rate (55.5% vs. 54.7%/56.2%) with modest per-item alignment (κ=0.34/0.28 vs. human A/B), while Gemini Spark collapses toward CLEAN (126/137; 92.0% CLEAN rate) and agrees with the humans essentially at chance (κ=0.03/−0.03; Table 11).

The taxonomy audit shows the same divergence plus a second failure mode: Gemini's 48-item taxonomy concentrates on clear_image_model_reasoning_failure (25/48) and never uses three of the eight classes, whereas GPT-5.6 distributes labels across all eight (exact-class agreement with humans: GPT-5.6 41.7%/37.5%, Gemini 31.2%/27.1%, human A–B 85.4%). MiMo's taxonomy collapse (43/48 to one class) and Gemini's two collapses (binary and tax-

Table 11. Binary clean/ambiguous audit agreement (n=137). Entries are exact agreement (%) with Cohen's κ in parentheses; CLEAN-rate column shows each auditor's base rate.

Auditor	Human A	Human B	GPT-5.6	Gemini	CLEAN
Human A	—	82.5 (0.65)	67.2 (0.34)	55.5 (0.03)	54.7
Human B		—	64.2 (0.28)	54.0 (−0.03)	56.2
GPT-5.6 Sol			—	57.7 (0.07)	55.5
Gemini Spark				—	92.0

onomy) are distinct failure modes of automated auditing: passing format checks—valid labels, all IDs present, no duplicates—is insufficient; audit pipelines need inter-rater agreement and label-distribution diagnostics. We therefore treat all LLM-generated ambiguity labels as sensitivity tools, not authoritative ground truth. Protocol caveats: the GPT-5.6 original interactive pass used contact-sheet batches (disclosed in its report) and was redone one-item-per-request as gpt56_clean_137.csv; the Gemini run is one-item-per-request (run run_gemini_spark_clean_137_001). Both are recorded as exploratory external evidence; the frozen v1 automated audit plus the two human re-audits remain the primary audit evidence in this paper.

Released resource. The layered multi-rater data — the automated audit, two independent human re-audits, and three LLM audit-reliability checks (MiMo-V2.5, GPT-5.6 Sol, Gemini Spark), together with all per-item agreement statistics — is versioned and released as a citable artifact under results/iaa/ (MANIFEST.json, ratings CSVs, agreement JSONs, and the blind-rating tooling).

7. Reproducibility and Artifact Index

Code: anonymized archive accompanying this submission (no external links). Every headline number in this paper maps to a committed artifact; the audit script scripts/make_paper_figures.py recomputes the main tables directly from the raw prediction CSVs (output archived at paper/fig/audit_output.txt).

Environments. VSR/7B and SITE runs: Ubuntu 24.04, Python 3.12.13, torch 2.12.1+cu130, transformers 5.14.1, datasets 5.0.1, peft 0.20.0, 1× RTX A6000 (48 GB). 2B runs: earlier python3.10-era environment (SmolVLM2 wrapper). MiMo-V2.5: public API (OpenCode Go gateway), thinking disabled, 12 parallel workers, per-request token accounting in results/mimo/usage_report.json (total spend reported there). No lock files exist; requirements.txt at the repository root declares the pinned top-level dependencies. Seeds recoverable from the archive: canonical VSR split seed 42 (dataset.train_test_split calls in the pipelines), IAA bootstrap seed 20260811 (scripts/compute_iaa.py), family-contrast bootstrap seed 20260812 (scripts/bootstrap_family_contrast.py), two-stage CV random_state=42. Training-run seeds were not recorded in the metrics JSONs (a known limitation); the prepared multi-seed extension (scripts/run_seed_variance.py / scripts/aggregate_seed_variance.py) records and reports per-run seeds explicitly.

Canonical freeze. The experimental result set is a frozen snapshot. Canonical tables are generated by scripts/build_canonical_tables.py; clean-label results by scripts/clean_label_orientation.py (automated-audit table) and scripts/build_human_clean_label_table.py (versioned human re-audit table); publication figures by scripts/make_publication_figures.py; SITE paired transfer statistics by scripts/compare_site_zeroshot_lora.py; VSR/SITE headline recomputation by scripts/make_paper_figures.py (output archived at paper/fig/audit_output.txt); MiMo-V2.5 evaluation and analysis by scripts/eval_mimo_vsr.py and scripts/mimo_analysis.py (predictions and summaries under results/mimo/).

Two-stage reproducibility caveat. The two-stage pipeline requires per-patch embeddings (results/probe/patch_embeddings.pkl, 8.2 GB, not committed). Regeneration command (GPU, ≥40 GB): python scripts/extract_patch_embeddings.py from the repository root, then python scripts/two_stage_reasoning.py; a SHA-256 checksum of the generated file will be recorded in the release (results/probe/patch_embeddings.pkl.sha256) once it is regenerated. The committed evidence is the aggregate results/two_stage_results.json (agreement-based, see App. D); scripts/two_stage_reasoning.py additionally scores per-example ground-truth accuracy and saves per-example decisions (results/two_stage_predictions.csv) when rerun, so the ground-truth variant is produced by the same regeneration step.

SITE secondary-subset audit note (corrected before submission). An earlier derived analysis (compare_site_zeroshot_lora.py v1, which fed the initial zeroshot_image_metrics.json-adjacent figures and the first canonical site table) reconstructed the secondary subset by applying ORIENTATION_KEYWORDS to the question text only, yielding n=1,824 (47.3% raw / 22.6% CAA). The frozen protocol's pre-specified definition — results/site/site_protocol.json → frozen_ids.secondary

WACV #*****

Headline number	Artifact
74.0 / 76.6 / 76.5 / 68.3 (2B)	smolvlm2_*_metrics_*.json
80.9 / 84.7 / 84.3 / 82.9 / 83.1 (7B)	*_metrics_*.json
62.8 / 63.5 / 65.7 / 66.4 (orient. family)	same + prediction CSVs
Per-relation cells (Table 1 (main))	orientation_analysis.json
48-failure taxonomy	orientation_persistent_*.json
Probe tables (App. 3)	probe/*.json
Two-stage (App. D, agreement-based rows 15–18)	two_stage_results.json
Consistency (Fig. 4 (main))	consistency_stats_all.json
Vision-side p-values	vision_side_comparison.json
SITE numbers	site/zeroshot_7b_predictions.csv site/zeroshot_image_metrics.json
MiMo-V2.5 VSR / consistency / audit (Table 1 row, verdict table, App. 3)	mimo/*.csv, mimo/mimo_vsr_summary.json
Human re-audits + inter-source agreement (App. 6)	iaa/rater2_*.csv, iaa/rater3_*.csv, iaa/iaa_results.json, iaa/consensu
LLM auditor reliability (MiMo / GPT-5.6 / Gemini; App. 6)	iaa/gpt56_clean_137.csv, iaa/gpt56_taxo_48.csv, iaa/gemini_clean_
Bootstrap family contrasts (§D)	bootstrap_family_contrast.json
MiMo format sensitivity (§A)	mimo/mimo_parseable_sensitivity.json

Table 12. Headline-number → artifact → commit index (all paths relative to results/).

(3,272 examples; 2,024 with images), intersected with the 2,591 evaluated IDs — is the authoritative subset, and all numbers in this paper use it (n=2,024; zero-shot 48.3% / 24.0% CAA; VSR-LoRA 48.7% / 24.5%, p=0.70). The derived question-only figure was corrected before submission; both definitions support the same conclusion (orientation ≪ primary), but only the frozen-ID set matches the pre-specification. No prediction CSV was modified and no model was rerun.

Data licenses/access. VSR: cambridgeltl/vsr_random (HF). SITE: frankyveteran/SITE-Bench, CC-BY-4.0 (media in 130 GB zips; requires HF read token). LoRA adapters are committed and loadable with peft; base-model weights via HF Hub.

Exact artifacts for the probe tables (App. 3): results/probe/probe_results.json, results/probe/patch_probe_results.json, results/probe/grounded_probe_results.json, results/probe/clear_subset_results.json, results/probe/grounded_boxes.json.

Runbook (one documented command per headline result). From the repository root, with the requirements.txt environment: canonical VSR tables python scripts/build_canonical_tables.py; headline recomputation + archive python scripts/make_paper_figures.py > paper/fig/audit_output.txt; automated clean-label table python scripts/clean_label_orientation.py; human re-audit tables python scripts/build_human_clean_label_table.py –rater rater2 and –rater rater3; inter-source agreement python scripts/compute_iaa.py (+ scripts/iaa_consensus_subsets.py); consistency tables python scripts/consistency_verdict_table.py and scripts/eval_consistency_flips.py; family contrasts python scripts/bootstrap_family_contrast.py; MiMo sensitivity python scripts/mimo_parseable_sensitivity.py; SITE transfer statistics python scripts/compare_site_zeroshot_lora.py; two-stage (requires the regenerated embeddings) python scripts/two_stage_reasoning.py. A CI-style smoke check runs the headline audit on a fresh clone; all prediction CSVs are committed, so the numerical results reproduce without any model or GPU.